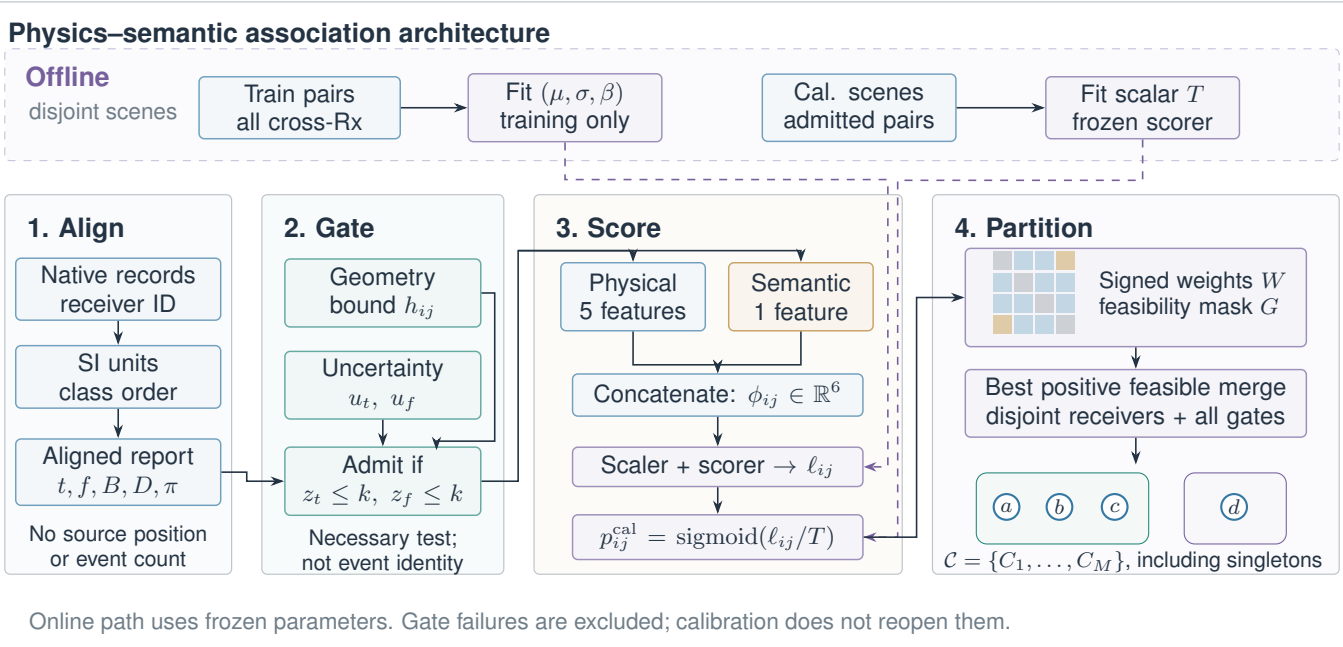

Physics–semantic association architecture
Offline
disjoint scenes
Train pairs
all cross-Rx
Fit (μ, σ, β)
training only
Cal. scenes
admitted pairs
Fit scalar T
frozen scorer
1. Align
Native records
receiver ID
SI units
class order
Aligned report
t, f, B, D, π
No source position
or event count
2. Gate
Geometry
bound h_{ij}
Uncertainty
$u_t,\ u_f$
Admit if
$z_t \le k,\ z_f \le k$
Necessary test;
not event identity
3. Score
Physical
5 features
Semantic
1 feature
Concatenate: $\phi_{ij} \in \mathbb{R}^6$
Scaler + scorer $\to \ell_{ij}$
$p_{ij}^{\mathrm{cal}} = \mathrm{sigmoid}(\ell_{ij}/T)$
4. Partition
Signed weights W
feasibility mask G
Best positive feasible merge
disjoint receivers + all gates
a
b
c
d
$\mathcal{C} = \{C_1, \ldots, C_M\}$, including singletons
Online path uses frozen parameters. Gate failures are excluded; calibration does not reopen them.